Green Home Score & Retrofit Decision Intelligence Engine

A Physics-Guided, Machine-Learned Architecture for Dual Homeowner and Lender Decarbonisation

Executive Abstract: Domestic heating and power account for over 20% of UK greenhouse gas emissions. The UK government mandates that all domestic properties reach EPC Band C by 2035, yet millions of homeowners face severe friction: fragmented contractor claims, opaque capital paybacks, and unexpected bills. Concurrently, mortgage lenders (such as Lloyds Banking Group) face statutory obligations to measure and reduce PCAF Scope 3 Category 15 financed emissions without exposing themselves to greenwashing liabilities. We present an end-to-end decision intelligence platform bridging this divide. The engine integrates: (1) an explainable 0–100 Green Home Score built upon 7 strictly monotonic sub-scores; (2) greybox thermal physics (SAP/RdSAP transmission heat-loss) coupled with calibrated ML surrogates; (3) a sequential retrofit simulator enforcing structural feasibility and per-fuel accounting; (4) hedonic property value uplift intervals (+2% to +5%); (5) interactive Thermal Twin customer & bank-level portfolio views; and (6) a zero-math LLM architecture with AST/regex guardrails ensuring 100% numerical groundedness (±1% tolerance).

1. Problem Statement & Domain Understanding

Domain Context & UK Regulatory Framework: Transforming the UK housing stock is hindered by four core domain complexities: (i) *Building Envelope Heterogeneity:* pre-1930 solid stone/brick walls, uninsulated suspended timber floors, and cavity-wall variants react differently to upgrades; (ii) *Interaction Non-Additivity:* thermal upgrades reduce heating load, making heat pumps more efficient but drastically reducing their gross kWh savings if computed independently; (iii) *Per-Fuel Split Dynamics:* electrification eliminates domestic gas demand (0.20275 kg CO_2e/kWh) while increasing grid electricity import (0.20733 kg CO_2e/kWh), which must be reported transparently; and (iv) *Mortgage Book Attribution:* lenders must accurately evaluate financed carbon emissions: *Financed Emissions* = *(Outstanding Loan Balance / Property Value)* × *Operational Carbon* under PCAF standard.

Pillar / Objective	Domain Complexity & Failure Modes	Engine Architectural Solution & Formal Invariant
1. Green Home Score	EPC ratings are static, step-wise (A–G), and conflate asset characteristics with occupancy behaviors.	Transparent formula: $S = \sum w_k s_k$ across 7 monotonic sub-scores. Fixed weights, explainable component points.
2. Poor Home Detection	Spacious detached homes consume high gross kWh yet may have superior thermal insulation, risking false-positive flags.	Normalised consumption intensity (kWh/m^2) + fabric health checks. Ranked by feasible % reduction in purchased grid energy.
3. Retrofit Feasibility	Recommending cavity insulation on solid walls or heat pumps in uninsulated homes causes mould, poor COP, and customer distrust.	Hard structural feasibility rules + customer tenure constraints. Strict sequential order: Fabric → Controls → Heating → Solar PV.
4. Energy & Carbon Cuts	Heat pump COP converts thermal energy into electrical power; blending fuels conceals increased electrical costs.	Per-fuel accounting: electrical surge reported as negative saving ($\Delta E_{elec} < 0$). Official DESNZ 2026 carbon factors applied per fuel.
5. Property Value Impact	Hedonic pricing is prone to regional and macro confounding; asserting causal price hikes breaches consumer finance rules.	Hedonic ML regression with counterfactual baseline comparisons, outputting strictly as an indicative range (+2% to +5%, never causal promise).
6. LLM Grounding & Ethics	LLMs hallucinate costs and savings, creating severe legal mis-selling risks for mortgage providers.	LLM restricted to natural language translation of structured JSON engine runs. Regex validator intercepts hallucinated numbers (±1% tolerance).

2. Mathematical Formulation & Explainable Scoring Model

The Explainable Green Home Score (0–100): The composite score is computed as a weighted combination of 7 monotonic sub-scores: $S = 0.25\, s_{FF} + 0.20\, s_{CI} + 0.15\, s_{INS} + 0.15\, s_{HEAT} + 0.10\, s_{GLAZ} + 0.10\, s_{REN} + 0.05\, s_{CTRL}$. Classifications: **Highly efficient** (80–100), **Good** (65–79), **Moderate** (50–64), **Poor** (30–49), **Very poor** (0–29). Sub-scores employ piecewise linear interpolation & verified lookup tables (config/score_mappings.yaml). Crucially, a superior building attribute never produces a lower score (mathematically verified monotonicity). Missing fields trigger explicit penalty flags rather than silent zeroing.

Key Architectural Standouts & Dual-Lens Innovations:

- **Thermal Twin SVG House Visualiser:** Interactive UI maps element heat loss (W/K) onto a cross-sectional house (roof, walls, glazing, floor). Homeowners dynamically toggle retrofits to see instant heat-loss rebalancing, score shifts, and utility bill updates in real time.
- **Lender Portfolio Decarbonisation View:** Provides Lloyds risk teams with bank-wide mortgage book analytics: PCAF Scope 3 Category 15 financed emissions, EPC score distributions, and a prioritized green lending target list ranked by greatest carbon reduction per £ invested (ΔtCO_2e / £).
- **Stateful Customer Memory Store:** Tracks multi-turn customer constraints (e.g., 'rented tenant', '£5,000 budget cap', 'conservation area'). The feasibility engine prunes infeasible measures across sessions while maintaining complete audit trails.
- **Deterministic Safety Sandbox & Verifier:** All scoring, physics simulations, and carbon calculations execute in deterministic Python. The LLM functions purely as an explanatory surface. Every response passes through an AST/regex grounding inspector before reaching the user.

3. High-Level Approach & Pipeline Architecture

The end-to-end architecture is built upon a modular, decoupled data and computation pipeline structured across five sequential stages:

Stage 1: Ingestion & Split	Stage 2: Scoring & Audit	Stage 3: Sequential Simulator	Stage 4: Carbon & Hedonic ML	Stage 5: Grounded AI & UI
Properties, smart meter bills, sales history, EPC data. **Customer Split:** 80/10/10 prevents data leakage.	**Green Home Score (0–100):** 7 sub-scores (weights 0.25–0.05). Identifies poor homes & fabric defects.	**Feasibility & Physics:** Fabric → Controls → Heat → Solar. Heat pump electricity surge as negative saving.	**DESNZ 2026:** Elec & Gas CO_2e factors. **Hedonic Regressor:** Indicative value uplift range (+2% to +5%).	**Grounded LLM:** Tools only, ±1% numeric verification. **UI:** Thermal Twin & Lender Portfolio view.

Core Technical Sub-Systems & Implementation Mechanics:

1. Data Calibration & Seasonal Annualisation: Ingests synthetic properties calibrated against NEED consumption statistics, UK HPI house price indices, and Met Office Heating Degree Days (HDD). Partial billing records are annualised using seasonal heating profiles rather than unweighted scaling, with explicit data quality flags for missing gas/electric meters.
2. Sequential Retrofit Simulator (Physics Greybox): Upgrades are applied sequentially to model physical dependencies: *Step 1: Fabric* (Loft, Wall, Glazing) reduces transmission heat loss $H = \sum(U_i \times A_i)$; *Step 2: Smart Controls* reduce heating runtime by 6–10%; *Step 3: Heating System* replaces boiler with heat pump (COP 3.2), converting remaining space heating demand into electrical power; *Step 4: Solar PV* generates renewable electricity (900 kWh/kWp), offsetting grid imports up to daytime base load. This strictly eliminates double-counted energy savings.
3. Per-Fuel Energy & DESNZ Carbon Accounting: Heat pump retrofits record electric increases as negative savings ($\Delta E_{elec} < 0$) while recording gas reductions ($\Delta E_{gas} > 0$). Carbon reduction is computed via official DESNZ 2026 factors: $\Delta C = \Delta E_{elec} \times 0.20733 + \Delta E_{gas} \times 0.20275$ *kg* CO_2e.
4. Financed Emissions Engine (PCAF Standard): Computes customer loan attribution factor *(Outstanding Mortgage / Property Value)* × *Home Carbon*, allowing risk managers to identify high-emission collateral clusters and evaluate green loan subsidies.
5. Hedonic Property Value Regression: Trains an ensemble regressor (XGBoost/LightGBM) on Land Registry Price Paid sales. Predicts counterfactual value before vs after retrofit holding macroeconomic features constant, reporting an indicative uplift interval (+2% to +5%).

4. Planned Testing Methodology & Quality Assurance Matrix

To ensure uncompromising reliability across commercial banking and consumer protection standards, we have established a 4-tier testing matrix:

Testing Tier	Test Target & Invariants Checked	Methodology & Pass/Fail Criteria	Evaluation Metric / Target
Tier 1: Unit & Invariant Tests	Score weights sum to 1.0; monotonicity of ratings/insulation; hard feasibility constraints (no cavity fill on solid brick); COP heat pump accounting.	Pytest automated test suite across all 7 sub-scores. Exhaustive property permutations across age bands, wall types, and heating fuels.	100% pass rate (45/45 active tests passing); zero mathematical or physics bugs.
Tier 2: ML & Physics Benchmarks	Energy consumption prediction (kWh) and hedonic property valuation (Price Paid counterfactuals).	Customer-stratified 80/10/10 held-out evaluation. Compared against naive mean baseline and pure-physics RdSAP benchmarks.	Energy $R^2 \geq 0.85$, RMSE < 850 kWh; Valuation MAPE < 7.5% across held-out test splits.
Tier 3: GenAI Grounding Audits	Numerical groundedness, scenario consistency, missing data handling, and absence of hallucinated savings/costs.	50 adversarial test cases (unsupported retrofits, missing gas meters, contradictory tenant prompts). Numeric extraction & JSON cross-check.	Groundedness ≥ 95%, Factual Accuracy ≥ 95%, Unsupported Claims < 2%.
Tier 4: Fairness & Bias Audits	Algorithmic fairness across property vintages (pre-1930 vs modern), low-value homes, and regional archetypes.	Subgroup error slicing: check whether low-income/older properties suffer higher error or receive financially infeasible advice.	Subgroup RMSE parity ratio > 0.80; prioritises grant-funded options for low-value bands.

5. Responsible AI, FCA Consumer Duty & Decarbonisation Governance

Regulatory Compliance & Anti-Greenwashing Safeguards: Under the UK Competition and Markets Authority (CMA) Green Claims Code and the Financial Conduct Authority (FCA) Consumer Duty, institutions face severe penalties for unsubstantiated environmental or financial projections. Our engine implements four structural governance safeguards: (1) *Strict Causal Disclaimers:* property-value uplifts are bounded as indicative observational correlations (+2% to +5%), explicitly disclaiming causal guarantees; (2) *Grant Prioritisation for Vulnerable Households:* low-income or EPC E/F/G homes automatically surface UK grant pathways (ECO4, Boiler Upgrade Scheme £7,500 heat pump grant); (3) *AST Output Verifier:* every numeric claim in conversational responses is cross-checked against the engine's structured JSON payload; discrepancies trigger immediate fallback; (4) *Auditability:* all 7 sub-score contributions, physics inputs, and counterfactuals are logged for regulator inspection.

Implementation Readiness & M2 Roadmap: Synthetic town data calibrated (10,000 homes, 235k monthly bills). Core schemas, seasonal annualisation, explainable scoring, DESNZ 2026 carbon calculation, sequential savings simulator, and FastAPI endpoints are fully operational (45 unit/integration tests green). Prior to Stage M2: (i) train final LightGBM/XGBoost models against NEED splits; (ii) execute the full 50-case adversarial AI benchmark; and (iii) assemble the customer Thermal Twin and Lender Portfolio views.